

MASTER OF BUSINESS ADMINISTRATION (MBA)

Examination Solutions: Management Information Systems & E-Governance (CC-105)

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BG-172/2

Subject: Management Information Systems & E-
Governance

Year: 2025

Time Allowed: 3 Hours

Session: 2024–2026

Full Marks: 70

GROUP — A (COMPULSORY QUESTIONS)

Question 1: Answer the following questions.

1 × 5 = 5 Marks

(a) Define system software.

System software refers to the foundational computer programs, such as operating systems, device drivers, and diagnostic utilities, designed to control and manage underlying hardware components and provide an isolated execution environment for application software.

(b) Define flowchart.

A flowchart is a standardized, visual representation that uses specialized geometric shapes and directional arrows to diagrammatic map the sequential steps, logic paths, and operational workflows of an algorithm or business process.

(c) What is data mining?

Data mining is the computer-assisted process of analyzing massive digital datasets through statistical algorithms, mathematical modeling, and machine learning to uncover hidden patterns, trends, and non-obvious relationships that support corporate decision-making.

(d) Define e-SCM.

Electronic Supply Chain Management (e-SCM) utilizes internet-based digital networks and integrated communication channels to plan, execute, and monitor end-to-end logistics, inventory distribution, supplier collaboration, and customer product fulfillment dynamically.

(e) Define OLAP.

Online Analytical Processing (OLAP) is a specialized software technology that empowers decision-makers to analyze multi-dimensional data across several attributes simultaneously, supporting high-speed corporate querying, trend analysis, and comprehensive business reporting.

Question 2: Define operating system. Write down about different functions of operating system.

5 Marks

An Operating System (OS) is the primary system software component that acts as an intermediary, secure interface between a computer's physical hardware components and the end-user or running application programs. Its main objective is to abstract complex low-level physical capabilities into manageable software services, ensuring that the computer infrastructure operates with maximum efficiency, reliable performance, and systematic stability.

The system accomplishes this goal through several core management functions. First, **Processor Management** schedules and coordinates how execution requests access the Central Processing Unit (CPU), using techniques like multitasking or multi-threading to balance performance across active tasks. Second, **Memory Management** dynamically tracks, allocates, and reclaims segments of main semiconductor memory (RAM) as applications load and close, ensuring programs operate smoothly without running out of resources. Third, **File System Management** organizes digital storage media into clear files and directory structures, controlling user read/write access permissions and keeping records safe from accidental loss.

Additionally, the operating system handles **Device Management** by using specialized software drivers to manage communication with external peripherals like printer networks, monitors, and storage arrays. Finally, it provides comprehensive **Security and Error Detection** features, including encrypted user login frameworks, firewalls, and system error alerts, which safeguard corporate data arrays from unauthorized access or software failures. Through these coordinated operations, the OS establishes the essential operational infrastructure that enables modern enterprise networks and applications to exist.

GROUP — B (LONG ANSWER QUESTIONS)

Question 3: Define management Information system. Briefly describe about Simon's framework of management information system.

15 Marks

A Management Information System (MIS) is an integrated, computer-based user-machine ecosystem designed to systematically capture, process, store, and distribute structured data from internal operations and external environments. This processed information is translated into meaningful business insights that support operational tracking, tactical management, and strategic decision-making within corporate enterprises. Far from being a simple collection of software or hardware components, an MIS coordinates technology infrastructure, standardized procedures, and human workflows to provide a single, consistent view of corporate truth. By automating routine data processing and delivering structured reports, an MIS breaks down departmental silos, improves organizational communication, and helps leaders respond quickly to changing market dynamics and regulatory frameworks.

To maximize the business value of an MIS, its structural data models must align with how corporate leaders naturally analyze and solve problems. This connection is best understood through Simon's framework of management decision-making, developed by Nobel Laureate Herbert A. Simon. This influential conceptual model views decision-making not as a single, sudden event, but as a multi-stage process split into three distinct, sequential phases: **Intelligence, Design, and Choice**.

The **Intelligence Phase** forms the initial foundation, focusing on scanning the internal and external environments to identify problems, inefficiencies, or emerging business opportunities. Within an MIS infrastructure, this phase relies heavily on continuous data monitoring systems, automated inventory tracking tools, and exceptional reporting modules. For example, if a manufacturing line's production output falls below a pre-set threshold or raw material costs rise unexpectedly, the MIS flags these variations, alerting managers that a decision is required.

Once a problem or opportunity is identified, the process moves into the **Design Phase**. This phase centers on formulating, developing, and analyzing alternative courses of action. Here, the MIS changes roles from a monitoring tool to an analytical engine. Managers leverage data modeling tools, forecasting algorithms, and "what-if" simulation modules to map out different solutions. For instance, if the intelligence phase identified a supply shortfall, the design phase might model three alternative strategies: contracting a new local supplier, increasing order volumes with an existing vendor for bulk discounts, or shifting production schedules to use alternative components. The system calculates the projected costs, implementation timelines, and operational risks for each path.

The final stage is the **Choice Phase**, where the decision-maker critically evaluates the structured alternatives generated during the design phase and selects the optimal path. The role of the MIS here is to provide clear decision support tools, such as multi-attribute comparison sheets or executive dashboards, that display the trade-offs of each choice side by side. After a choice is made, modern iterations of Simon's framework add an **Implementation and Review Phase** to track real-world results against the original forecasts, feeding that performance data back into the system to refine future decision cycles.

In conclusion, Simon's framework provides a vital blueprint for building modern information systems. By designing software that matches these cognitive phases, organizations ensure their information tools actively assist managers through every stage of a problem, transforming raw data into systematic, strategic corporate action.

Question 4: What do you mean by Functional Information System? Briefly describe about Personnel and HR Information system?

15 Marks

A Functional Information System is a specialized enterprise information architecture designed to support a specific business function, operational domain, or department within an organization. Unlike broad, cross-functional systems like Enterprise Resource Planning (ERP) platforms that link the entire company together, functional information systems tailor their database structures, data processing models, and reporting layouts to the unique operational and strategic requirements of individual departments. Typical business examples include Financial Information Systems, Marketing Information Systems, Production/Operations Information Systems, and Human Resource Information Systems. By gathering deep, specialized data within a single department, these systems streamline daily transactions, optimize local resource use, and provide functional managers with targeted reports to track performance against specific business metrics.

Among these functional frameworks, a **Personnel and Human Resource Information System (HRIS)** is a critical tool for managing an organization's most valuable asset: its human capital. Modern organizations use an HRIS to track, manage, and analyze employee data throughout the entire employment lifecycle, from initial recruitment to retirement. A robust HRIS operates across operational, tactical, and strategic management tiers to handle several core responsibilities.

At the operational level, an HRIS automates repetitive administrative tasks, lowering costs and minimizing data errors. Key capabilities include **Time and Attendance Management**, which uses biometric check-ins or digital logs to track employee hours, shift rotations, and leave requests. This module connects directly with the **Payroll Processing System**, which automatically calculates gross salaries, tax withholdings, medical insurance deductions, and compliance reporting. By digitizing these tasks, companies eliminate manual errors and ensure employees are paid accurately and on time.

At the tactical level, the system provides management with tools to track development and optimize performance. **Talent Acquisition and Tracking** systems (Applicant Tracking Systems or ATS) manage online job descriptions, organize incoming resumes, coordinate interview rounds, and streamline onboarding workflows for new hires. Alongside recruitment, the **Performance Management** module records quarterly evaluations, tracks professional goals, and identifies skill gaps. This data feeds into **Training and Development Subsystems**, which manage corporate training schedules, track mandatory certifications, and ensure the workforce maintains the skills needed to meet corporate standards.

At the strategic level, an HRIS provides senior executives with advanced **Workforce Analytics and Planning** tools. By evaluating historical patterns in data, the system generates deep insights into employee turnover rates, tracking which departments experience the highest attrition and why. It also models future retirement pipelines, helping executives plan leadership succession and manage recruitment budgets years in advance. These analytics allow companies to transition from reactive hiring to proactive talent strategies, aligning long-term staffing levels with broader corporate growth goals.

In conclusion, a Personnel and HR Information System transforms human resource management from an administrative overhead function into a strategic advantage. By organizing employee data within a secure, functional database, an HRIS ensures operational accuracy, builds a stronger talent pipeline, and helps corporate

leadership make evidence-based decisions that maximize workforce productivity, engagement, and retention across the enterprise.

Question 5: Describe about data communication. Write down about components of data communication and different communication channels.

15 Marks

Data communication refers to the systematic transmission, exchange, and receipt of digital data between two or more autonomous computing devices across a shared communication medium. This process allows modern organizations to distribute information seamlessly, power global networks like the internet, and support interconnected business operations. To ensure data travels accurately and securely without corruption or loss, data communication relies on a structured blend of hardware components and standardized software protocols. These protocols establish clear rules for encoding, transmitting, and decoding digital signals across different network infrastructures.

To understand how data communication works, we must first look at its five fundamental components, which form the core architecture of any network path:

- **The Message:** This is the actual data or information that needs to be transmitted. It can take various digital formats, including plain text, numbers, images, audio files, or streaming video.
- **The Sender:** The source device that creates and initiates the data transmission. Typical examples include personal computers, mobile workstations, network servers, or security cameras.
- **The Receiver:** The destination device intended to receive and process the transmitted message. This can be another computer, a printer, a television display, or a mainframe database server.
- **The Transmission Medium:** The physical channel or path that connects sender and receiver, acting as the highway over which data signals travel.
- **The Protocol:** A binding set of rules, syntax specifications, and synchronization procedures that govern how data is transmitted. Without protocols, two connected devices might send signals but would be unable to interpret each other's data correctly.

The transmission medium itself is broadly divided into two major categories of communication channels: ****Guided (Wired) Media**** and ****Unguided (Wireless) Media****, each offering distinct advantages in terms of speed, bandwidth capacity, and physical deployment costs.

****Guided Media**** uses physical cables to direct data signals along a specific path. The most common type is ****Twisted-Pair Cable****, which consists of color-coded copper wires twisted together to reduce electromagnetic interference. It is widely used in local area networks (LANs) and telephone systems due to its low cost and flexibility. ****Coaxial Cable**** features a central copper conductor surrounded by a thick layer of insulation and shielding, providing better protection against external noise. This makes it ideal for cable television networks and high-bandwidth internet connections. The most advanced guided channel is ****Fiber-Optic Cable****, which transmits data as pulses of light through thin strands of glass or plastic. Fiber optics offer massive bandwidth capacities, extremely high data transmission speeds, and total immunity to electromagnetic interference, forming the foundational backbone of modern global telecommunication networks.

****Unguided Media****, on the other hand, transmits electromagnetic signals through space or air without a physical wire, providing the flexibility needed for mobile communication. ****Radio Waves**** travel in all directions

and pass easily through physical obstacles, making them perfect for local AM/FM broadcasting, cellular phone networks, and standard Wi-Fi setups. **Microwaves** are unidirectional signals that travel in a straight line, requiring a clear line-of-sight between transmission towers. They are commonly used for long-distance terrestrial links and connecting separate corporate buildings. Finally, **Satellite Communication** uses geosynchronous satellites orbiting the Earth to relay microwave signals over massive distances, providing internet connectivity, weather tracking, and broadcasting services to remote regions across the globe.

In summary, data communication is a vital foundation for modern business and technology. By combining reliable components with the right mix of wired and wireless channels, organizations can build secure, scalable networks that keep teams connected and power global digital operations.

Question 6: Briefly describe about Software Development Life Cycle (SDLC).

15 Marks

The Software Development Life Cycle (SDLC) is a structured, phased methodology and engineering framework used by software organizations to plan, design, develop, test, deploy, and maintain high-quality software applications. The primary goal of the SDLC is to deliver robust, dependable systems that meet or exceed user expectations while keeping development timelines and budgets firmly under control. By breaking down complex development projects into distinct, manageable stages with clear deliverables, the SDLC minimizes project risks, standardizes team communication, and provides management with tangible benchmarks to evaluate progress. This systematic approach ensures that final software products are stable, secure, and ready to scale within an enterprise environment.

A standard SDLC moves through a series of sequential, highly coordinated phases, each producing key documentation or artifacts that serve as inputs for the next stage:

- **Requirement Gathering and Analysis:** This foundational phase involves collaborating closely with clients, business analysts, and end-users to document exactly what the software needs to accomplish. The final requirements are recorded in a **Software Requirement Specification (SRS)** document, which serves as a binding blueprint for the entire project.
- **System Design:** Architects and senior developers use the SRS document to design the application's structural plan. This includes defining the hardware layout, choosing database models, mapping data paths, and designing user interfaces. This phase produces a **Design Specification Document (DSD)** that guides the development team.
- **Implementation or Coding:** This is the phase where the actual software is built. Software developers write clean, standardized code based on the design blueprints, using chosen programming languages (such as Java or C++) and frameworks. Teams often divide responsibilities into modules to keep development organized and maintainable.
- **Testing:** Once the code is written, it is handed over to quality assurance (QA) teams. Testers run the software in isolated test environments to find and track functional bugs, security vulnerabilities, and performance issues. They use various methods, including unit testing, integration testing, and User Acceptance Testing (UAT), to ensure the software matches the original SRS guidelines.
- **Deployment:** After the application passes all testing stages and is approved by stakeholders, it is officially released into the production environment. This step can occur in phases, such as a limited beta launch to a small group of users, or as a full-scale deployment across the entire organization.
- **Maintenance:** The cycle continues long after launch. During the maintenance phase, support teams monitor the live application to fix unexpected user issues, apply security updates, and implement feature enhancements based on real-world feedback, keeping the software useful and secure over time.

Depending on a project's scope, budget, and stability requirements, development teams choose from several SDLC methodologies. The classic **Waterfall Model** follows a strict, linear approach where each phase must finish completely before the next begins. This model works best for well-defined, predictable projects where requirements are unlikely to change mid-development. In contrast, modern **Agile Methodologies** focus on

flexibility and continuous improvement. Agile breaks projects down into small, iterative development blocks called sprints, allowing teams to deliver working software quickly and adapt to changing user feedback on the fly. Other popular frameworks include the **Spiral Model**, which focuses on rigorous risk analysis for large-scale engineering projects, and the **V-Model**, which pairs every development phase with a matching testing stage to ensure maximum code quality.

In conclusion, the Software Development Life Cycle is an indispensable management tool for modern software engineering. By enforcing clear documentation standards, structural milestone reviews, and disciplined testing phases, the SDLC provides a predictable roadmap that helps organizations deliver stable, scalable, and cost-effective software solutions that drive real business value.

Question 7: Define E-Governance. Write down about evolution, scope and current global trends of E-Governance.

15 Marks

Electronic Governance, universally known as E-Governance, is the systematic application of Information and Communication Technologies (ICT)—including internet networks, cloud infrastructure, mobile applications, and centralized databases—to transform how public administration operates. The primary objective of E-Governance is to enhance the transparency, accountability, and efficiency of public services, making them more citizen-centric and accessible. Rather than simply replacing paper workflows with digital screens, true E-Governance re-engineers administrative procedures to eliminate bureaucratic corruption, lower transactional overhead costs, and build a more inclusive relationship between the state and its citizens.

The **evolution** of E-Governance generally moves through four distinct structural phases as a nation's digital capacity grows. The first phase is **Information (Computerization)**, where governments set up basic web portals to publish static public information, such as regulatory policies, application forms, and office hours, replacing printed leaflets. The second phase is **Interaction**, which adds early two-way communication channels, allowing citizens to download forms, send emails to public departments, or submit basic digital queries. The third phase is **Transaction**, marking a major step forward by enabling citizens to complete full service workflows online, such as paying utility bills, filing income taxes, and renewing passports through secure digital payment gateways. The final phase is **Transformation**, where different government databases are completely integrated into a single network, allowing public services to share data instantly and provide seamless, automated support without requiring citizens to submit paper documents between separate departments.

The **scope** of E-Governance is exceptionally broad, spanning four primary interactions across public administration:

- **Government-to-Citizen (G2C):** Delivers direct public services to the population, including online birth certificate applications, digital land registration, and direct welfare transfers, making services accessible regardless of physical distance.
- **Government-to-Business (G2B):** Streamlines interactions with the private sector through automated tax filings, digital corporate registration platforms, and transparent e-procurement portals for public tenders, encouraging clean business investment.
- **Government-to-Government (G2G):** Enables seamless data sharing and collaboration between different ministries, state departments, and local authorities, reducing administrative friction and speeding up national policy planning.
- **Government-to-Employee (G2E):** Manages internal public sector workflows, including digital payroll systems, automated performance tracking, and online training portals for government workers.

Looking at **current global trends**, E-Governance is moving toward highly advanced, intelligent digital systems. A major trend is the implementation of **Cloud-First Policies** and **Decentralized Data Exchanges**, modeled closely after Estonia's secure X-Road infrastructure. This model allows different government databases to exchange data instantly while maintaining strict user privacy. Another key trend is the integration of **Artificial Intelligence (AI) and Predictive Analytics** to automate routine public requests, optimize urban traffic flows, and

detect financial fraud in public spending. Governments are also focusing heavily on **Mobile-First Governance**, building comprehensive smartphone applications that put public services directly in citizens' hands. Finally, as states shift more operations online, there is a massive global emphasis on strengthening **Cybersecurity and Data Protection Frameworks** like the EU's GDPR, ensuring that as public administration digitizes, citizen data remains safe, resilient, and trusted.

Question 8: Write short notes on any two of the following: (a) Make or Buy decision, (b) e-CRM, (c) Decision Support System, (d) Cloud Computing.

15 Marks

*Note: As per the examination format, comprehensive academic notes for any two options—**(c) Decision Support System** and **(d) Cloud Computing**—have been completed below to thoroughly satisfy the 650-750 words requirement.*

(c) Decision Support System (DSS)

A Decision Support System (DSS) is an interactive, computer-based information platform that combines data analytical models, operational databases, and user-friendly interfaces to help managers navigate semi-structured or unstructured decision-making problems. Unlike Transaction Processing Systems (TPS) that focus on recording routine, daily operations, a DSS is built specifically to assist middle and upper management when business conditions are unpredictable and rules are not fully predefined. A DSS does not replace human judgment; instead, it empowers decision-makers to test different scenarios and evaluate options through advanced analytical tools like "what-if" models, sensitivity analysis, and multi-variable optimization calculations. By pulling data from both internal systems and external market feeds, a DSS allows managers to explore complex business problems, simulate outcomes, and confidently choose the most effective corporate strategy.

The structural framework of a typical DSS relies on three interconnected core components that drive its analytical capabilities. The first is the **Data Management Subsystem**, which contains a database of relevant internal transaction data and external market intelligence, such as competitor pricing or industry economic indicators. The second is the **Model Management Subsystem**, which houses the mathematical, financial, and statistical models used to analyze the data, such as linear programming tools or trend forecasting modules. The third is the **User Interface Subsystem**, which provides an intuitive, graphical dashboard that allows non-technical managers to easily manipulate data models, run simulations, and view results through clear charts and tables. By combining these tools, a DSS serves as a vital corporate asset, enabling data-driven decisions that minimize risks, optimize resource use, and drive long-term business growth.

(d) Cloud Computing

Cloud Computing refers to the on-demand, internet-based delivery of computing services—including servers, data storage repositories, databases, networking infrastructure, software applications, and analytical tools—under a flexible pay-as-you-go pricing model. Instead of buying, building, and maintaining expensive on-premise physical datacenters, companies rent access to these computing resources from major third-party cloud providers like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform. This model shifts IT spending from heavy upfront capital expenditures (CapEx) to predictable operational expenses (OpEx), allowing companies to access enterprise-grade infrastructure with minimal initial investment and absolute scalability.

The architecture of cloud computing is broadly categorized by three core service models and three primary deployment strategies. The service models include **Infrastructure as a Service (IaaS)**, which rents basic hardware resources like virtual servers and storage; **Platform as a Service (PaaS)**, which provides a pre-configured environment for developers to build and test software applications; and **Software as a Service (SaaS)**, which delivers fully functional web applications over the internet, such as Microsoft 365 or Salesforce.

In terms of deployment, companies choose between **Public Cloud** (sharing infrastructure managed by a provider), **Private Cloud** (building dedicated cloud environments for maximum security), or **Hybrid Cloud** (linking public and private systems to balance flexibility with data control). By providing global data availability, automatic software updates, and built-in disaster recovery features, cloud computing serves as a powerful engine for corporate agility, digital transformation, and scalable business innovation.